MAC283 HOMEWORK - WEEK 1

Ralph Cange

ralph.cange@live.lagcc.cuny.edu 9/16/23

1. **Briefly describe what an ALU (Arithmetic-Logic Unit) is. (in ~100 Words)**

The CPU's internal circuit known as the ALU processes data by performing arithmetic and logic operations. While its logic operations comprise of and, or, and not operations, it will execute fundamental mathematical operations like addition, subtraction, multiplication, and division. Bits, which are binary digits used in digital and computational communication, are subjected to these procedures. There are many logical circuits inside the ALU referred to as half adders and full adders. The full adder will generate the sum of three inputs and the carry value, which moves to the next bit, while the half adder will only produce the sum of two inputs.

2. **What is the Von Neumann Bottleneck? Briefly describe solutions to the Bottleneck. (in ~100 Words)**

Main memory, a CPU or processor, and a link connecting the memory and the CPU make up the Von Neumann Bottleneck design. Each of the places in main memory has the capacity to store binary values for both data and instructions. Within the confines of the computer's system clock cycles, instructions are carried out sequentially. The clock speed quantifies how many cycles your CPU performs per second in gigahertz.

3. **If a store sells two "1GB" storage disks, both for $100.00 but one disk is a 10^9 and the other disk is 2^{30} - Which is a better deal and why?**

Despite having almost 1GB of capacity each, the disk with 230 only has 107374182 bytes, while the disk with 109 only has 10000000 bytes. Because it has slightly higher storage capacity than the other disks, disk 230 will be more valuable.

4. Perform the following base conversions using subtraction or division-remainder:

45810 = ___**12122**_____ $_{3}$

66710 = ___10132_____ $_{5}$

151810 = ___4266____ $_{7}$

440110 = __6030_____ $_{9}$

5. Perform the following base conversions:

201013 = ____172_____ $_{10}$

32025 = ___427_____ $_{10}$

16056 =____437____ $_{10}$

6879 = ___565_____ $_{10}$

6. Convert the following decimal fractions to binary with a maximum of six places to the right of the binary point:

24.78125 =11000.11001

193.03125 =11000001.00001

291.796875 =100100011.11001

17.1240234375 =10001.0011

7. Convert the hexadecimal numbers to binary.

$BC13_{16}$ =1011100000010011

$7B02_{16}$ =0111101100000010

DEAD $BEEF_{16}$ =1101111010101101

8. Represent the following decimal numbers in binary using 8-bit signed magnitude.

47 = 00101111

139 = 10001011

–25 = 11100111

–108 = 10010100